CMU Voice Chat Server Setup

For the server maintainer. Voice chat runs inside the normal game server, so there is no separate voice server to host.

What you need

- A Linux server running Ubuntu 22.04 or 24.04, with root (sudo) access.
- The server package, `SS14.Server_linux-x64.zip` (from us, or see "Building the package" at the end).
- The setup script, `setup-server.sh` (in the repo at `Content.CMU/Tools/LiveServer/`).
- Access to our domain's DNS settings. Browsers only allow microphone access on HTTPS pages, so the voice page needs a real domain name, not just an IP.

Step 1: Point a subdomain at the server

1. In the DNS settings for our domain, add an A record. For example: name `voice`, value = the server's public IP. This gives `voice.ourdomain.com`.
2. If the domain is on Cloudflare, set that record to "DNS only" (grey cloud, not orange).
3. Wait until it resolves. `ping voice.ourdomain.com` should show the server's IP.

Any subdomain name works. Use it everywhere this guide says `voice.ourdomain.com`.

Step 2: Upload the files

From the computer that has the zip and the script:

```
scp SS14.Server_linux-x64.zip setup-server.sh root@SERVER_IP:~/
```

Step 3: Run the setup script

```
ssh root@SERVER_IP
sudo bash setup-server.sh voice.ourdomain.com "CMU Main"
```

Replace `voice.ourdomain.com` with the subdomain from Step 1 and `"CMU Main"` with the server name you want.

The script does the following:

- Installs the .NET 10 runtime and Caddy (Caddy handles HTTPS and gets the certificate by itself).

- Opens the firewall for SSH, 80, 443 and 1212 (TCP and UDP). Port 1213 stays closed on purpose.
- Installs the game to `/opt/cmu-server` and runs it as the `cmu-server` service.
- Sets up `https://voice.ourdomain.com/voice/`.

If your hosting provider has its own firewall in their web panel, open the same ports there too.

Step 4: Check that it works

1. Open `https://voice.ourdomain.com/voice/` in a browser. You should see a padlock and a box asking for a code.
2. Connect to the game at `voice.ourdomain.com:1212`.
3. In game, press Esc and click Voice Chat. The page opens with your code filled in and says "Connected as (your name)".
4. Allow the microphone. Hold CapsLock to talk, or J to talk on your headset radio.

Updating the server

Upload the new zip and run the same command from Step 3 again. It replaces the game files and keeps your existing `server_config.toml`.

Useful commands

What	Command
Watch the server log	`journalctl -u cmu-server -f`
Restart the server	`systemctl restart cmu-server`
Watch the Caddy (HTTPS) log	`journalctl -u caddy -f`
Edit server settings	`nano /opt/cmu-server/server_config.toml`, then restart

Common problems

Problem	Fix
Certificate error on the voice page	The subdomain isn't pointing at the server yet, it's proxied through Cloudflare (orange cloud), or ports 80/443 are blocked (check the provider's firewall panel). Look at the Caddy log.
Page loads but never connects	The game server isn't running. Check the server log.
"That code didn't work"	Codes reset when a player leaves. Press Voice Chat in game again for a new one.
Microphone blocked	The page has to be opened through the https domain address, not the raw IP.

Already running the server your own way?

You don't have to use the script. Voice chat only needs these settings in `server_config.toml`:

```
[cmu.voice]
enabled = true
ws_bind = "127.0.0.1:1213"
public_url = "https://your.domain/voice/"
```

Then put an HTTPS reverse proxy in front of it. With Caddy:

```
your.domain {
        handle /voice/ws* {
                reverse_proxy 127.0.0.1:1213
        }
        handle /voice* {
                reverse_proxy 127.0.0.1:1212
        }
        handle {
                respond 404
        }
}
```

If you use nginx instead, `/voice/ws` must be proxied as a WebSocket (Upgrade and Connection headers). If the proxy runs on a different machine, set `trusted_proxies` under `[cmu.voice]` to its IP and change `ws_bind` so it can reach port 1213.

Optional settings

All under `[cmu.voice]`. Defaults are fine for most servers.

Setting	Default	What it does
`range`	10	Normal talking range in tiles
`radio_enabled`	true	Allow talking over headsets
`radio_common`	true	Allow voice on the Common channel
`max_connections_per_ip`	6	Limit on voice pages per IP
`telephone_enabled`	true	Voice carries over phone calls
`megaphone_range`	20	Range in tiles when holding a megaphone

Admins can mute a player's voice with the `voicemute` command, or give a lasting voice ban from the chat ban menu (Voice box).

Building the package

On a Windows PC with the repo checked out, run this from the repo folder:

```
powershell -ExecutionPolicy Bypass -File Content.CMU\Tools\LiveServer\build-server.ps1
```

The zip ends up at `release\SS14.Server_linux-x64.zip`.